GradientCIO

MARKETABLE ALTERNATIVES PORTFOLIO CONSTRUCTION

Northstar Hedge Fund Portfolio

Portfolio construction review · prepared for the Investment Committee

AS OF
2026-06-30

TOTAL NAV
$1,000.0M

PORTFOLIO WEIGHT
12.0%

POLICY BENCHMARK
Northstar Absolute Return Index

01 Executive Decision

Redemption terms need confirmation before funding

RECOMMENDATION

We recommend retaining the 12.0% marketable-alternatives allocation while shifting 2.0 percentage points from equity long/short to macro and relative-value roles S3 S4.
Committee action requested: approve the target structure and staged manager review, subject to verified redemption terms and completed diligence S3 S7 S8.

MARKETABLE-ALTERNATIVES WEIGHT	ACTIVE WEIGHT	RELATIVE RETURN	LIQUID SHARE
12.0%	0.0pp	+0.6pp	8.0%
Policy target 12.0% S3	Total-fund target S3	Trailing 1Y versus benchmark S5	Monthly or better as % NAV S3

Recommendation and action requested

We recommend a 2.0 percentage-point strategy shift within the existing 12.0% allocation. Committee action requested: approve the target mix and authorize manager review, with no funding until redemption and diligence conditions are complete S3 S4 S7 S8.

DECISION SUMMARY

DECISION ELEMENT	PROPOSED ACTION	EVIDENCE	STATUS
Total allocation	Hold at 12.0%	S3	Compliant
Equity long/short	Reduce from 5.0% to 3.0%	S3 S4	Proposed
Macro and relative value	Increase from 4.0% to 6.0%	S3 S4	Proposed

Conditions and limits

Confirm notice periods, gates and side-pocket terms for each affected fund and complete ODD before implementation S7 S8.

Key judgment — Redemption terms

Observation: Notice periods, gates and side-pocket terms are unconfirmed for the affected funds S1. Why it matters: Funding before confirmation could lock capital beyond policy limits. Uncertainty: Terms come from manager documents still in review. What would change the view: Confirmed terms inside the policy liquidity window.

Key judgment — Sleeve size

Observation: Marketable alternatives are 12.0% of total NAV S2. Why it matters: The 2.0pp strategy shift stays within this allocation. Uncertainty: Hedge fund values are reported with a lag. What would change the view: A revised valuation that moves the weight outside its range.

Key judgment — Liquidity

Observation: Only 8.0% of the sleeve is liquid within the policy window S3. Why it matters: Redemption terms, not returns, constrain how fast the shift can happen. Uncertainty: Gates can be applied at the manager's discretion. What would change the view: A manager applying or lifting a gate.

02 Mandate, Benchmark and Data Basis

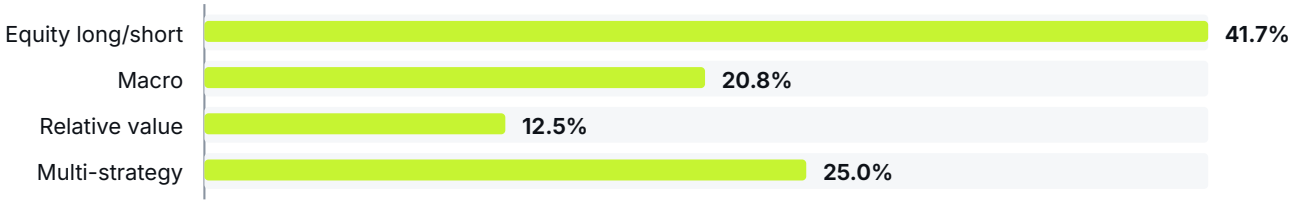
Portfolio	**Northstar Total Fund** S2
Mandate	**Diversifying hedge-fund portfolio** S8
Benchmark	**Northstar Absolute Return Index** S5
Horizon	**5 years** S6
Return basis	**Net** S5
Data scope	**Illustrative sample data** S1 S1

GOVERNED FRAMEWORK

AREA	OBJECTIVE OR LIMIT	OBSERVED	STATUS	SOURCE
Marketable alternatives	8.0%–16.0%	12.0%	Compliant	S3
Single manager	At or below 4.0% of NAV	4.0%	Near limit	S3 S4
Monthly-or-better liquidity	At least 7.0% of NAV	8.0%	Compliant	S3

03 Current Hedge Fund Portfolio

HEDGE-FUND STRATEGY ALLOCATION



CURRENT SLEEVE

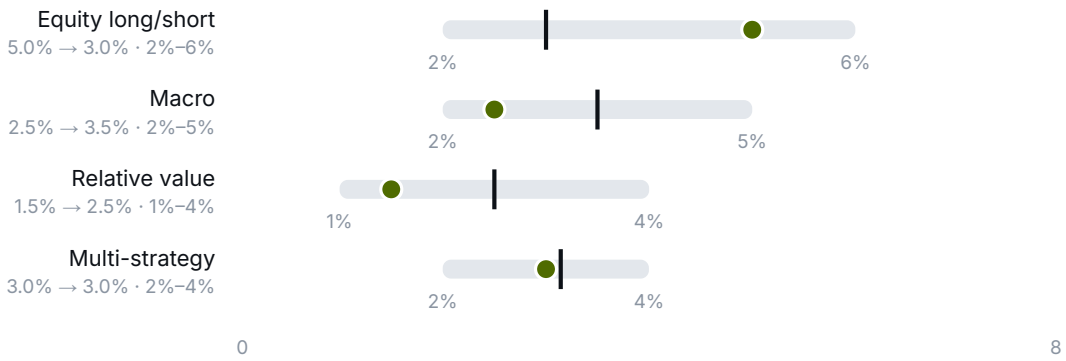
STRATEGY / MANAGER	ROLE	VALUE	% PORTFOLIO	% SLEEVE	AS OF
Example Alpha Partners	Equity long/short	$50.0M	5.0%	41.7%	2026-06-30 S4
Example Macro Fund	Macro	$25.0M	2.5%	20.8%	2026-06-30 S4
Example Relative Value Fund	Relative value	$15.0M	1.5%	12.5%	2026-06-30 S4
Example Multi-Strategy Fund	Multi-strategy	$30.0M	3.0%	25.0%	2026-06-30 S4

Strategy and role labels are shown as returned; they do not establish leverage, liquidity or underlying positions S4.

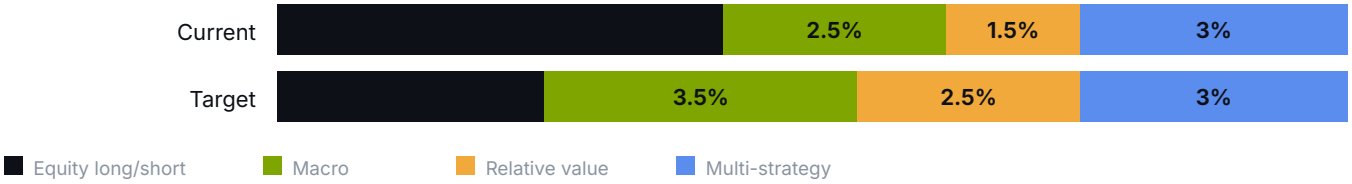
04 Target Strategy and Manager Structure

A broader strategy mix lowers single-role concentration

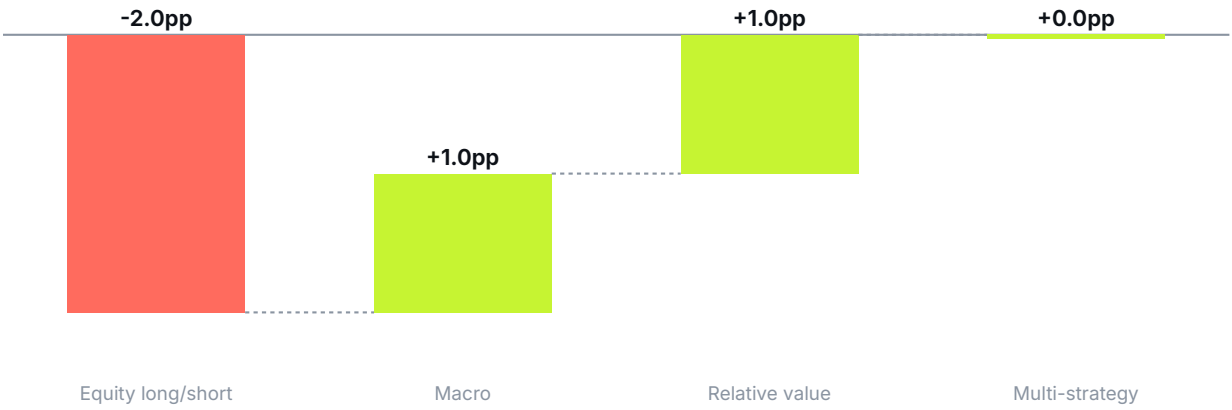
CURRENT WEIGHT VERSUS POLICY RANGE



CURRENT AND TARGET STRATEGY MIX



TARGET ALLOCATION CHANGES (PERCENTAGE POINTS)



CURRENT VERSUS TARGET

STRATEGY / ROLE	CURRENT	TARGET	CHANGE	RANGE / CONSTRAINT	RATIONALE
Equity long/short	5.0%	3.0%	-2.0pp	2.0%–6.0%	Reduce largest strategy concentration S3 S4
Macro	2.5%	3.5%	+1.0pp	2.0%–5.0%	Increase diversifying role subject to diligence S7 S8
Relative value	1.5%	2.5%	+1.0pp	1.0%–4.0%	Broaden strategy mix subject to terms S7 S8
Multi-strategy	3.0%	3.0%	0.0pp	2.0%–4.0%	Maintain current role S3 S4

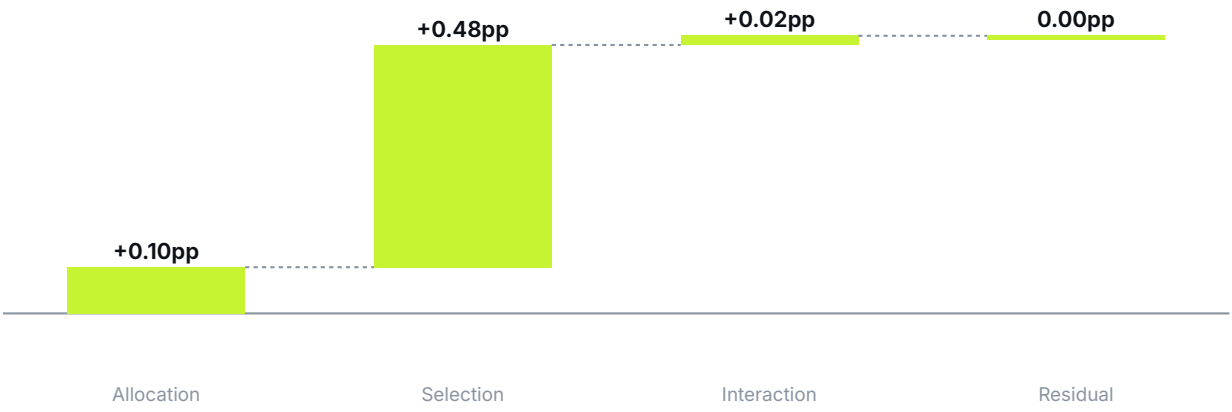
Analysis — Why this target

Observation: The target reallocates 2.0 percentage points from equity long/short to macro and relative value S3 S4. Why it matters: The largest strategy concentration declines under the target. Uncertainty: CMA, regime, findings and term coverage remain incomplete S7 S8. What would change the view: Completed records contradicting either proposed strategy role S7 S8.

The target is a committee proposal and is not a saved portfolio change S3 S8.

05 Historical Performance and Attribution

GOVERNED ATTRIBUTION EFFECTS



Returned attribution effects; residual is shown explicitly S5.

STANDARD PERIODS

PERIOD	PORTFOLIO / SLEEVE	BENCHMARK	EXCESS	COVERAGE
1Y hedge-fund sleeve	6.4%	5.8%	+0.6pp	Complete S5
3Y hedge-fund sleeve (ann.)	5.9%	5.5%	+0.4pp	Complete S5

ATTRIBUTION EFFECTS

EFFECT	CONTRIBUTION	SOURCE
Allocation	+0.10pp	S5
Selection	+0.48pp	S5
Interaction	+0.02pp	S5
Residual	0.00pp	S5

Governed attribution method and reconciliation metadata follow S5.

Method	**Brinson-Fachler** S5
Linking	**Symmetric Carino** S5
Basis	**Net, monthly linked** S5
Tolerance	**1 bp; reconciled** S5

06 Factor, Currency and Concentration Evidence

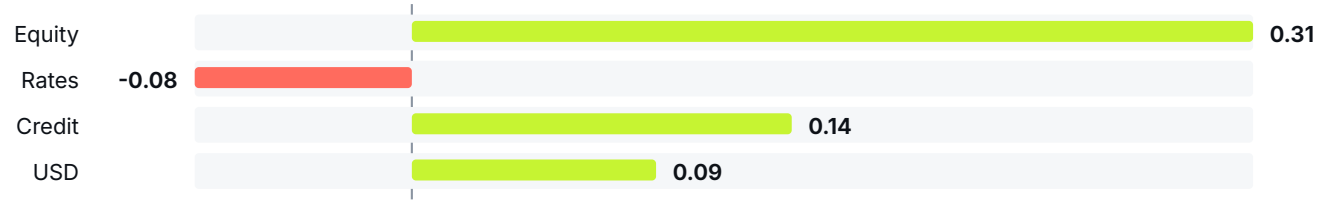
Returned exposures show where diversification remains incomplete

STRATEGY-CORRELATION MATRIX

	EQUITY L/S	MACRO	RELATIVE VALUE	MULTI-STRATEGY
EQUITY L/S	1.00	0.18	0.42	0.51
MACRO	0.18	1.00	0.12	0.24
RELATIVE VALUE	0.42	0.12	1.00	0.47
MULTI-STRATEGY	0.51	0.24	0.47	1.00

Returned strategy-correlation evidence S7.

MODEL-IMPLIED FACTOR EXPOSURES



GOVERNED CONCENTRATION

AXIS	OBSERVED	LIMIT	STATUS	SOURCE
Largest manager	4.0% of NAV	4.0%	**Near limit**	S3 S4
Largest strategy	5.0% of NAV	6.0%	**Compliant**	S3 S4
13F equity subset	Not available — manager identifiers incomplete	Not assessed	**Not assessed**	S7

Analysis — Manager limit is binding

Observation: The largest manager is at the 4.0% policy limit S3 S4. Why it matters: New funding should not increase that concentration. Uncertainty: Model factors and incomplete look-through omit the full hedge-fund book S7. What would change the view: Complete exposure records showing concentration below the 4.0% threshold S4 S7.

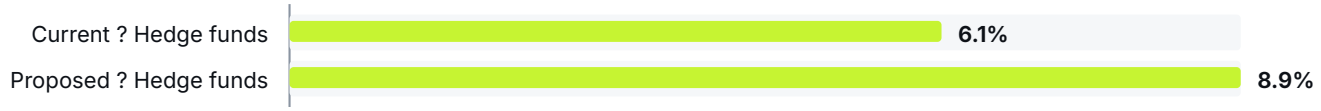
Look-through limitation

Any Form 13F evidence is lagged, long-only US-listed equity, excludes shorts, derivatives, cash and non-US listings, and is not a complete hedge-fund exposure view S7.

07 Forward Return and Risk

Governed assumptions frame tradeoffs rather than forecasts

FORWARD EXPECTED-RETURN COMPARISON



Governed forward assumptions; assumptions, not forecasts S6.

Assumption set	**2026 Strategic CMA** S6
Regime	**Baseline** S6
Horizon	**5 years** S6
Currency	**USD** S6
Context fingerprint	**test-ma-forward-2026** S6

MARKETABLE-ALTERNATIVES FORWARD ASSUMPTIONS

CLASS	EXPECTED RETURN	EXPECTED VOLATILITY
Hedge funds	6.1%	8.9%
Absolute return	5.5%	7.1%

> **Analysis — Assumptions show tradeoff**
> Observation: Hedge funds carry higher expected return and volatility than absolute return in the selected assumptions S6. Why it matters: The target should preserve role diversification, not maximize one measure. Uncertainty: Class assumptions are not manager-level forecasts. What would change the view: Refreshed assumptions reversing the relative return or volatility results S6.

These figures are assumptions under the selected context, not forecasts or guarantees S6.

08 Liquidity, Redemption and Operational Terms

LIQUIDITY COMPOSITION



LIQUIDITY PROFILE

BUCKET / TERM	% NAV	AMOUNT	REQUIREMENT	STATUS
Monthly or better	8.0%	$80.0M	At least 7.0%	**Compliant**
Quarterly	4.0%	$40.0M	Monitor	**Available**

REDEMPTION EVIDENCE

MANAGER / FUND	FREQUENCY	NOTICE	GATE / SIDE POCKET	SOURCE
Example Alpha Partners	Quarterly	60 days	Not available — terms not returned	S8
Example Macro Fund	Monthly	30 days	No gate reported	S8
Example Relative Value Fund	Monthly	45 days	Not available — terms not returned	S8
Example Multi-Strategy Fund	Quarterly	90 days	Not available — terms not returned	S8

Redemption frequency and notice are user-document evidence; missing gates, side pockets and leverage remain open items S8.

09 Manager Lineup and Diligence

CURRENT AND PROPOSED ROLES

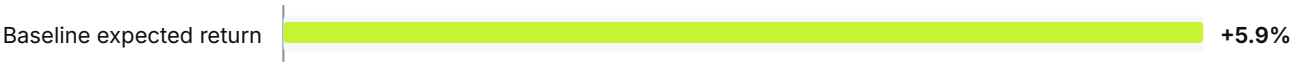
MANAGER	STRATEGY / ROLE	EXPOSURE	DILIGENCE STATUS	PROPOSED ACTION	NEXT STEP
Example Alpha Partners	Equity long/short	$50.0M	Current; 1 Medium finding S7	Redeem $20.0M	Confirm notice and gate
Example Macro Fund	Macro	$25.0M	Current S7	Add $10.0M	Refresh ODD
Proposed Relative Value Manager	Relative value	Not funded	Not assessed S7	Consider $10.0M	Run comparison and ODD

Diligence condition

No proposed manager may be funded until comparison, operational diligence and performance-integrity review are complete S7.

10 Scenarios and Robustness

GOVERNED SCENARIO RESULTS



Saved-portfolio stress is unavailable S8.

SUPPORTED SCENARIOS

SCENARIO	RETURN / LOSS	RISK MEASURE	BASIS	SOURCE
Baseline assumptions	5.9% expected return	8.1% volatility	Proposed sleeve	S6
Saved-portfolio stress	Not available — no governed saved-portfolio stress result	Not available	Unavailable	S8

> **Selected-series sandbox — not saved-portfolio analytics**
> Not available — no Strategy Lab return series selected S8.

11 Implementation Plan

IMPLEMENTATION STEPS

PHASE	ACTION	SIZE	FUNDING SOURCE	CONDITION	OWNER
1	Submit conditional equity long/short redemption	2.0pp	Existing sleeve	Terms confirmed	Investment staff
2	Fund macro role	1.0pp	Redemption proceeds	ODD refresh complete	CIO
3	Fund relative-value role	1.0pp	Redemption proceeds	Manager approved	Committee

ALTERNATIVES CONSIDERED

ALTERNATIVE	BENEFIT	COST / RISK	REASON NOT SELECTED
No change	No transition	Concentration remains	Does not address watch item S3 S4
Immediate full rebalance	Faster target alignment	Terms and costs unavailable	Conditions incomplete S7 S8

12 Risks, Open Items and Approvals

DECISION RISKS

PRIORITY	RISK	EVIDENCE	MITIGATION / CONDITION	OWNER
High	Redemption restrictions	Gate terms unavailable S8	Verify legal terms before notice	Operations
Medium	Manager concentration	4.0% policy watch S3 S4	Stage redemption	CIO
Medium	Proposed manager diligence	Not assessed S7	Complete comparison and ODD	Investment staff

OPEN ITEMS AND APPROVALS

TIMING	ITEM	REQUIRED EVIDENCE	OWNER	STATUS
Before approval	Confirm redemption terms	Manager documents	Operations	Open S8
Before approval	Select relative-value manager	Comparison and ODD	Investment staff	Open S7
Next review	Refresh attribution	Governed return period	Performance team	Open S5

13 Coverage

Portfolio and allocation	**Available**	Illustrative sample data S2
Policy	**Available**	Allocation, liquidity and concentration S3
Exposure	**Available**	Manager market values S4
Returns and attribution	**Incomplete**	Five-year sleeve period unavailable S5
Forward assumptions	**Available**	2026 CMA S6
Diligence and look-through	**Incomplete**	Proposed manager and 13F subset incomplete S7
Redemption terms and scenarios	**Not available**	Gate terms and saved-portfolio stress not returned S8

A Appendix — Sources and methods

Based on 8 of 9 evidence areas.

TAG	EVIDENCE	SOURCE	AS OF	CHECK
S1	Organization and capabilities	list_organizations; get_gradient_capabilities	2026-06-30	**Passed**
S2	Portfolio and allocation	list_portfolios; get_portfolio_structure	2026-06-30	**Passed**
S3	Policy assessment	check_portfolio_policy	2026-06-30	**Passed**
S4	Hedge-fund exposure	get_portfolio_exposure	2026-06-30	**Passed**
S5	Returns and attribution	get_portfolio_historical_returns; get_portfolio_attribution	2026-06-30	**Incomplete**
S6	Forward assumptions	get_capital_market_assumptions	2026-06-15	**Passed**
S7	Diligence and look-through	get_portfolio_exposure	2026-06-30	**Incomplete**
S8	Mandate, redemption terms and gaps	get_gradient_capabilities	2026-06-30	**Advisory**

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

SOURCE AUDIT TRAIL

TAG	KEY PARAMETERS	DIGEST
S1	organization=test	`sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa`
S2	allocation_tree	`sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa`
S3	portfolio=test	`sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa`
S4	all pages	`sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa`
S5	all sections	`sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa`
S6	baseline	`sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa`
S7	illustrative	`sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa`
S8	illustrative	`sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa`

Data scope for every source: Illustrative sample data. Each digest is the SHA-256 fingerprint of the exact data the report used.

B Appendix — Server Metric Methods and Disclosures

SERVER METRIC METHODS

SOURCE	QUANTITY	METHOD / FORMULA	VERSION	BASIS / UNITS / TOLERANCE
S3	Policy status	Governed assessment	portfolio-policy-v1	Returned status
S4	Exposure	Server exposure aggregation	portfolio-exposure-v1	USD market value
S5	Returns and attribution	Linked returns; Brinson-Fachler	portfolio-attribution-v1	Net; 1 bp tolerance

- Forward values are assumptions, not forecasts or guarantees S6.
- Form 13F is lagged, long-only and incomplete for hedge funds S7.
- Redemption, gate and side-pocket terms require source documents S8.
- Portfolio Analytics and Strategy Lab are separate analytical scopes.
- This AI-assisted construction analysis is not investment, legal or compliance advice.